

Statistical modeling

Module 02

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Further than correlation and dependence measures

Correlations and dependence measures

- ▶ Quantify the dependence.
- ▶ Do not qualify the type of dependence.
- ▶ Are limited to pairwise dependence.
- ▶ Are not useful for prediction.

We now build models



where

- ▶ the dependence of y on x is explicit and can be used for prediction,
- ▶ x can be several variables, $x_1, \dots, x_p$,
- ▶ the uncertainty is quantified.

Variables in the study

There are two types of variables:

- ▶ Variables $x_1, \dots, x_p$, called features (machine learning context), independent variables, factors or covariates¹.
- ▶ A response y , also called outcome or dependent variable.

The response variable has a central role. It is a variable of interest either

- ▶ to be predicted.
- ▶ to study the relation between y and each features in x .

In the following, we use x for all the features, and detail $x_1, \dots, x_p$ when needed.

¹and other names in other scientific fields...

Examples

Interested in answering the above research question, (Willerman, et al, 1991) studied a sample of $n = 38$ college students²:

- ▶ Response (y): Performance IQ scores (PIQ) from the revised Wechsler Adult Intelligence Scale. This variable served as the investigator's measure of the individual's intelligence.
- ▶ Potential predictor (x_1): Brain size based on the count obtained from MRI scans (given as count/10'000).
- ▶ Potential predictor (x_2): Height in inches.
- ▶ Potential predictor (x_3): Weight in pounds.

²source: <https://onlinecourses.science.psu.edu/stat501/node/284>

Examples

(Colby, et al, 1987) wanted to find out if swallows, which live in underground burrows, also alter how they breathe. In an underground burrow, they varied the percentage of oxygen at four different levels (13%, 15%, 17%, 19%) and the percentage of carbon dioxide at five different levels (0%, 3%, 4.5%, 6%, 9%). Under each of the resulting $5 \times 4 = 20$ experimental conditions, the researchers observed the total volume of air breathed per minute for each of 6 nestling bank swallows. They obtained a sample of $n = 120$ nestling bank swallows³:

- ▶ Response (y): percentage increase in "minute ventilation," (Vent), i.e., total volume of air breathed per minute.
- ▶ Potential predictor (x1): percentage of oxygen (O2) in the air the baby birds breathe.
- ▶ Potential predictor (x2): percentage of carbon dioxide (CO2) in the air the baby birds breathe.

³source: <https://onlinecourses.science.psu.edu/stat501/node/285>

Examples

Example: In a company the weekly sales of ice-cream y is dependent on the advertising budget x_1 , the average temperature x_2 , and the holiday season x_3 . Past data are used to forecast the future sales in order to manage the supply/production processes.

Example: In a large professional association, a study about equality in employee salaries y is performed. The independent variables are the gender x_1 , the company sector x_2 , the education level x_3 , the seniority x_4 , the position in the company x_5 , and the hierarchy in the company x_6 . The impact of the gender on the salary (at same level of the other variables) is of interest to determine if anti-discrimination actions should be taken.

The model

To qualify/quantify the impact of x on y , a model is built

$$y = f(x_1, \dots, x_p) + e.$$

where

- ▶ function $f(x_1, \dots, x_p)$ represents what are the expected values y for given values of x .
- ▶ e is the error term of the model, expected to be 0. The error helps to describe the uncertainty of the model.

The function f needs to be estimated in order to give a good description of the outcome y for given x . The category of available functions is very large. The most used is linear functions, which gives the well known regression models which is the central topic of this course.

The data

The model is calibrated (estimated, more later) in order to match a sample of observations. Except specified differently, the notation is

- ▶ i denotes the observation label (e.g. subject, week, case, etc.).
- ▶ n is the sample size: number of observations.
- ▶ (y_i, x_i) or $(y_i, x_{i1}, \dots, x_{ip})$ the observed values (response and features) for observation i .

The observations are assumed to be independent given x .

Example: in the study on college students, i is a student label, $n = 38$, and for example, for subject 1, we have

$$y_1 = 124, x_{11} = 81.69, x_{12} = 64.5, x_{13} = 118.$$

Subjects 1 and 2 are independent: knowing y_1 does not bring any extra information on y_2 if x_2 is known.

Regression

Regression is a model where the function f is assumed to be a linear combination of the variables x .

Assuming that we have n observations $\{y_i, x_{i1}, \dots, x_{ip}\}$, then the model is

$$y_i = b_0 + b_1x_{i1} + \dots + b_px_{ip} + e_i,$$

where

- ▶ $b_0, b_1, \dots, b_p$ are the coefficients
- ▶ e_i is the residual error

$$e_i = y_i - (b_0 + b_1x_{i1} + \dots + b_px_{ip}).$$

Note: for simplicity, we dropped the dependence on b . To be "super precise" we could write $y(x; b)$ and $e(x; b)$.

Regression

Given a set of coefficients $b = (b_0, \dots, b_p)$, for a subject i having characteristics $\{y_i, x_{1i}, \dots, x_{pi}\}$, the prediction of y_i is

$$y(x_i) = b_0 + b_1 x_{1i} + \dots + b_p x_{pi}$$

The estimated error is $e(x_i) = y_i - y(x_i)$. It is expected to be 0.

Several names are given to these models:

- ▶ regression,
- ▶ linear regression,
- ▶ linear model, general linear model (not to be confounded to generalized linear models),
- ▶ multiple regression (as opposed to simple regression, because there are more than one x).

Regression: illustration

In a population of children, seasonal pocket money (y in \$) is studied as a function of the age (x_1 in yrs) and the average grand-parent age (x_2 in yrs).

Model:

$$y = 307.71 + 10.94x_1 + 2.85x_2 + e$$

i	y_i	x_{1i}	x_{2i}	$y(x_i)$	$e(x_i)$
1	565	10	50	559.6	5.39
2	601	13	60	620.9	-19.9
3	560	14	40	574.9	-14.9
...	...	...	...	...	...

Regression: interpretation of the coefficients

Coefficient b_j is the average increase of y for an increase of 1 unit of x_j . Sometimes they are called the sensitivity of y on x_j .

Example:

- ▶ Pocket money increases in average by \$10.94 when age increases by 1 year,
- ▶ Pocket money increases in average by \$2.85 when the average grand-parent age increases by 1 year.

Regression: out-of-sample prediction

New subject prediction: y is unknown, $x = (x_1, \dots, x_p)$ is given,

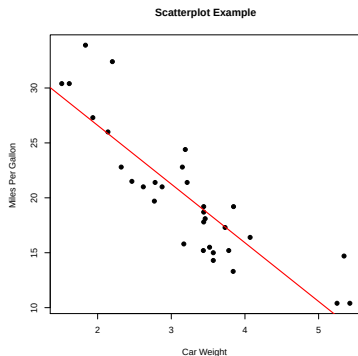
$$y(x) = b_0 + b_1x_1 + \dots + b_px_p$$

Example: for pocket money, a new 8 year-old child (y unknown) with 55 year-old grand-parents is predicted:

$$y = 307.71 + 10.94 \times 8 + 2.85 \times 55 = 551.98$$

Regression: graphical representation

For the simple regression (i.e. $p = 1$), use scatter plot and add the regression line (red). Below, a car consumption y (Miles per Gallon) as a function of its weight x_1 .



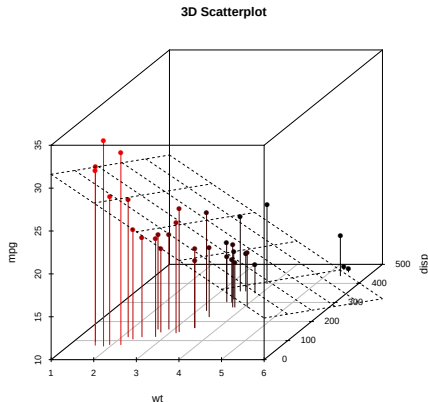
$$\text{Model: } y = 37.3 - 5.3x_1 + e$$

Regression: graphical representation

With $p = 2$, one could use a regression plan (not often in practice).

Below consumption y as function of the weight x_1 and the displacement x_2 .

$$y = 35.0 - 3.35x_1 - 0.018x_2 + e$$



Estimation of a regression: OLS

There are thus $p + 2$ parameters to estimate: $(b_0, \dots, b_p)$ and σ . From a sample $\{y_i, x_1, \dots, x_p\}_{i=1}^n$, what are the best coefficients $b = (b_0, \dots, b_p)$ to use?

Minimize the sum of squares:

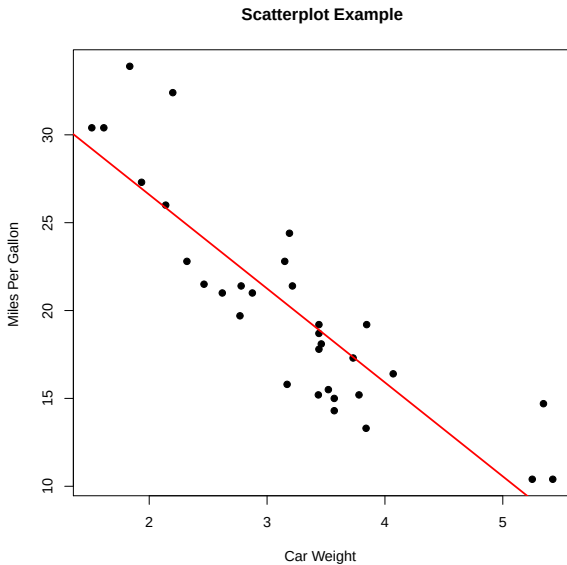
$$\begin{aligned} SS(b) &= \sum_{i=1}^n e(x_i)^2 = \sum_{i=1}^n \{y_i - y(x_i)\}^2 \\ &= \sum_{i=1}^n (y_i - b_0 - b_1 x_{i1} - \dots - b_p x_{ip})^2 \end{aligned}$$

Let $\hat{b}$ be the solution,

$$\hat{b} = \arg \min_b SS(b)$$

This is the Ordinary Least Squares (OLS).

OLS: illustration



OLS: calculations

Compute the derivatives and equalize to 0:

$$\left. \frac{\partial SS(b)}{\partial b} \right|_{b=\hat{b}} = 0.$$

For b_0 :

$$\frac{\partial SS(b)}{\partial b_0} = -2 \sum_{i=1}^n (y_i - b_0 - b_1 x_{i1} - \dots - b_p x_{ip}).$$

For $b_j, j = 1, \dots, p$:

$$\frac{\partial SS(b)}{\partial b_j} = -2 \sum_{i=1}^n x_{ij} (y_i - b_0 - b_1 x_{i1} - \dots - b_p x_{ip}).$$

OLS: calculations

The estimating equations for b are

$$\sum_{i=1}^n x_{ij}(y_i - \hat{b}_0 - \hat{b}_1 x_{i1} - \cdots - \hat{b}_p x_{ip}) = 0, \quad j = 0, \dots, p.$$

where $x_{i0} = 1$ for all $i = 1, \dots, n$.

Finding $\hat{b}$ by solving these estimating equations can be done using iterative numerical algorithms. A convenient way of representing (and solving) the system is by matrix representation.

Matrix notations

$$\begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} 1 & x_{11} & \cdots & x_{1p} \\ \vdots & \vdots & \vdots & \vdots \\ 1 & x_{n1} & \cdots & x_{np} \end{pmatrix} \begin{pmatrix} b_0 \\ \vdots \\ b_p \end{pmatrix} + \begin{pmatrix} e_1 \\ \vdots \\ e_n \end{pmatrix},$$

$$y = Xb + e$$

Sum of squares:

$$SS(b) = (y - Xb)^T (y - Xb) = e^T e = \|e\|^2$$

Estimating equations:

$$X^T (y - X\hat{b}) = 0$$

Ordinary Least Squares estimate:

$$\hat{b} = (X^T X)^{-1} X^T y$$

OLS: predictions and errors

OLS predictions:

$$\hat{y}_i = \hat{b}_0 + \hat{b}_1 x_{i1} + \cdots + \hat{b}_p x_{ip},$$

OLS errors:

$$\hat{e}_i = y_i - \hat{y}_i.$$

In matrix notations:

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\mathbf{b}}, \quad \hat{\mathbf{e}} = \mathbf{y} - \hat{\mathbf{y}}$$

Example

A real estate firm builds a selling price prediction model for a house (y in \$1'000) using Sq feet (x_1), Age (x_2), Bedrooms (x_3), Bathrooms (x_4), # cars in the garage (x_5). They use a data base with historical prices ($n = 250$). Using the OLS, they find

$$\text{Price} = 31.1 + 63.1x_1 - 1.14x_2 - 8.4x_3 + 3.5x_4 + 28.2x_5 + e$$

Could you explain the negative coefficient $\hat{b}_3$?

Example:

For the IQ study, when calling the R function `lm`, the OLS is computed and extracted when calling the `summary` function:

```
> mod <- lm(PIQ ~ Brain + Height + Weight, data=IQ)
> summary(mod)
[...]
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	1.114e+02	6.297e+01	1.768	0.085979	.
Brain	2.060e+00	5.634e-01	3.657	0.000856	***
Height	-2.732e+00	1.229e+00	-2.222	0.033034	*
Weight	5.599e-04	1.971e-01	0.003	0.997750	

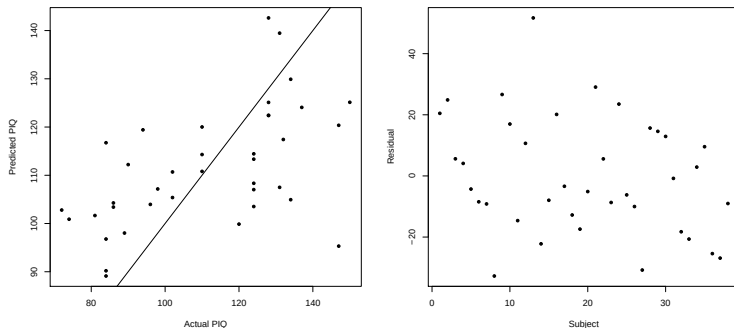
```
---
[...]
```

The fitted model is

$$y = 111.4 + 1.060x_1 - 2.732x_2 + 0.00056x_3 + e.$$

Example:

The (in-sample) predicted responses against the actual values, $\hat{y}$ vs y and the residuals ($\hat{e}$) are shown below



Conclusion: from the left-hand plot we see that the prediction is not very good.

Sum of squares

Sum of squares are important quantities to qualify the model quality (goodness-of-fit) and much more.

Residual Sum of squares (e stands for **errors**):

$$SS_e = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = (y - \hat{y})^T (y - \hat{y}) = \hat{e}^T \hat{e}.$$

Total Sum of squares:

$$SS_T = \sum_{i=1}^n (y_i - \bar{y})^2$$

Sum of squares of the **Regression**:

$$SS_R = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$$

Sum of squares decomposition

It can be shown that

$$SS_T = SS_R + SS_e$$

Interpretation:

- ▶ A perfect regression predict y very well, i.e. $y_i \approx \hat{y}_i$ and thus

$$SS_e \approx 0, \quad SS_T \approx SS_R.$$

- ▶ A large SS_R indicates a good model fit to the data,
- ▶ A large SS_e indicates a bad model fit to the data.

Coefficient of determination: R^2

It is an indicator of the model quality. It is defined as the proportion of variation in y explained by the variation of x

$$R^2 = \frac{SS_R}{SS_T} = \frac{SS_T - SS_e}{SS_T} = 1 - \frac{SS_e}{SS_T}$$

- ▶ Because $0 \leq SS_R \leq SS_T$, $0 < R^2 < 1$,
- ▶ A large* R^2 indicates a good model fit to the data.

*How much large? please don't ask...

R^2 : Example

In the IQ example,

```
> summary(mod)
```

```
[...]
```

```
Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	1.114e+02	6.297e+01	1.768	0.085979	.
Brain	2.060e+00	5.634e-01	3.657	0.000856	***
Height	-2.732e+00	1.229e+00	-2.222	0.033034	*
Weight	5.599e-04	1.971e-01	0.003	0.997750	

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 19.79 on 34 degrees of freedom
```

```
Multiple R-squared:  0.2949,    Adjusted R-squared:  0.2327
```

```
F-statistic: 4.741 on 3 and 34 DF,  p-value: 0.007215
```

The R^2 is equal to 0.295.

R^2 : Example

In the real estate example, they found

$$SS_T = 1.27 \times 10^6, SS_e = 2.34 \times 10^5, SS_R = 1.04 \times 10^6.$$

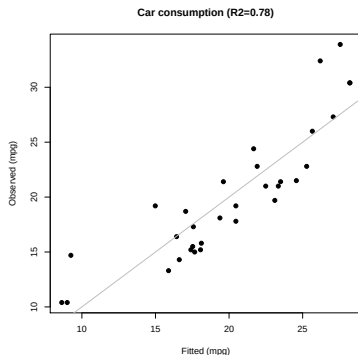
thus

$$R^2 = 1.04/1.27 = 0.82.$$

Thus, 82% of the variation of the price is explained by the variation of x_1, x_2, x_3, x_4, x_5 .

Graphical inspection of the model quality

The coefficient of determination R^2 is an indicator of model adequacy with the data. Often used in complement to the scatter plot of y against $\hat{y}$ (below for the car consumption example):



A good fit = points should be close to the (0,1) line (in grey)

Adjusted R^2

The adjusted R^2 is an alternative to the R^2 that accounts for the number of parameters in the model,

$$R_a^2 = 1 - \frac{SS_e/(n - p - 1)}{SS_T/(n - 1)}.$$

Suppose that two linear models achieve the same sum of squares SS_T and SS_e , and thus the same R^2 , then the model with the least number of parameters (p smaller) will have a higher R_a^2 .

The adjusted R^2 penalizes more complex models (more parameters). This principle is central in model selection (more later).

R^2 : Example

In the IQ example,

```
> summary(mod)
[...]
```

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	1.114e+02	6.297e+01	1.768	0.085979	.
Brain	2.060e+00	5.634e-01	3.657	0.000856	***
Height	-2.732e+00	1.229e+00	-2.222	0.033034	*
Weight	5.599e-04	1.971e-01	0.003	0.997750	

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 19.79 on 34 degrees of freedom
Multiple R-squared:  0.2949,    Adjusted R-squared:  0.2327
F-statistic: 4.741 on 3 and 34 DF,  p-value: 0.007215
```

The adjusted R^2 is equal to 0.2327. This is still bad...

R^2 : Example

In the real estate example, they found

$$SS_T = 1.27 \times 10^6, \quad SS_e = 2.34 \times 10^5, \quad SS_R = 1.04 \times 10^6.$$

thus

$$R_a^2 = 1 - \frac{2.34/(250 - 5 - 1)}{12.7/(250 - 1)} = 0.812.$$

We see that, when $p \ll n$ (here $5 \ll 250$), the R^2 and the adjusted R^2 are very close.